

Kunal Maheshwari

Mount Pleasant, MI, USA | (989) 906-2108 | kunal.maheshwari.work@gmail.com | <https://www.linkedin.com/in/kunalmaheshwari26>

SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of professional experience as a Software Developer and Data Engineer, specializing in building scalable microservices and distributed data pipelines. Leverages expertise in Java, Python, Spark, SQL, AWS, and Azure, coupled with experience in CI/CD automation and graduate-level work on large-scale data platforms and privacy-preserving AI applications, including Large Language Models (LLMs). Seeking to apply advanced skills in Backend Development and Software Engineering to a Software Engineer LMTS (Big Data Focus) role.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Designed and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime.
- Engineered CI/CD pipelines with Jenkins and Azure DevOps for continuous delivery across Azure Cloud and Oracle WebLogic environments, reducing deployment times by 30 percent.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring, reducing issue resolution time by 20 percent.
- Led a significant EDX platform upgrade, streamlining application architecture and dependency management for 50 percent faster delivery cycles.
- Collaborated in Agile and Scrum teams to deliver FHIR-compliant healthcare solutions using Git, SVN, and JUnit testing, ensuring high code quality.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered robust Java and EJB-based enterprise applications and RESTful web services for automotive clients.
- Achieved a 90 percent system performance improvement through SQL query optimization, database tuning, and modernization of legacy Java code.
- Implemented CI/CD pipelines using Jenkins and GitLab with automated testing and quality gates, enhancing release efficiency.
- Mentored junior developers through code reviews and debugging sessions, promoting clean coding practices and effective Software Engineering.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage for critical modules.
- Supported release planning and sprint execution using JIRA within Agile development teams, facilitating efficient Backend Development.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time, leveraging PySpark and Trino for efficient querying.
- Engineered an interactive Flask web application for data exploration with dynamic filtering and visualization, integrating Tableau for advanced analytics.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and utilized Kafka for real-time data streaming.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent and enhancing Large Language Models (LLMs) data preparation.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight for scalable data warehousing.

Privacy-Preserving AI Email Client (Capstone Project) (T5, LoRA, Python) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, utilizing Hugging Face libraries.
- Applied parameter-efficient fine-tuning using LoRA for low-resource training under privacy constraints, demonstrating expertise in Large Language Models (LLMs).
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, supporting functional testing with Jsunit.
- Explored backend development possibilities with Agentforce integration for enhanced email management and CRM functionalities.

SKILLS

- Programming Languages: Java, Python, SQL.

- Frameworks and Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, Jsunit.
- Web Services: REST API, SOAP API.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform, Kafka, Trino.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/ CD, Microservices, Large- scale data platforms, Infrastructure automation, Privacy- preserving LLM- based applications, Functional Testing, Software Engineering, Backend Development, CRM.
- Data Analysis and Visualization: Tableau, QuickSight.